

Abraham Miller

Senior Software Engineer in Life Sciences

+1 509 995 9071

abraham.r.m@gmail.com

www.abemill.com

Senior full-stack engineer with 6 years building enterprise-scale life sciences software. Expertise in lab automation, cloud services, and LIMS. AI tooling evangelist driving up to 10% productivity gains across teams. Published researcher in biology, applied math, and ML.

Experience

Senior Software Engineer, HighRes Biosolutions

January 2025 – Present

- Serve as key technical contributor for partnership with one of the largest global pharmaceutical companies, integrating their LIMS with our cloud laboratory orchestration and automation. At times performed urgent data remediation and system troubleshooting to maintain multi-million dollar scientific workflows under high-pressure timelines.
- Developed a reusable integration platform that enables connection between our laboratory orchestration services and diverse customer systems with abstractions that measurably reduce custom development across 3 separate client projects by ~30% compared to prior work.
- Led developer acceleration initiatives through AI tooling workshops and documentation, sharing productivity gains in company-wide meetings via live-demo full codebase refactors across 10k+ line projects using Claude Code.
- Enhanced team communication standards by proposing, then implementing and championing policies around structured Jira task management and Microsoft Teams usage, reducing notification noise and increasing team focus and output.

Software Engineer I → II, Ginkgo Bioworks, Inc.

September 2020 – August 2024 (4 years)

- Developed essential software functionalities including a Strain lineage browser tracking >100k cell lines, and designed data models for LIMS representing millions of complex lab samples for thousands of daily power users.
- Created, merged, and deployed >600 Pull Requests across 21 coding projects, frontend+backend using AWS, Kubernetes, Docker, Python, Django, React, TypeScript, Ruby on Rails, GraphQL, SQL, and other. Generally, technologically agnostic work.
- Provided reliable on-call support, resolving >800 user issues and saving >\$200k through averted downtime in certain memorable instances.
- Initiated an internal AI experimentation blog engaging >50 employees, resulting in 3-10% productivity gains for participants.

Software Engineer Contract, Lutz Lab, University of Washington

March 2020 – September 2020 (6 months)

- Single-handedly produced software-guided workflows and a footpedal-operated lab kiosk for a portable low-cost COVID-19 and HIV diagnostic system, achieving 12X efficiency gain over standard workflow operation. System has been deployed in Kenya and Nigeria.

Education

University of Washington, Seattle

BS/BA Double Degree, Computer Science & Philosophy

September 2015 – March 2020

Published Research

Simpler and faster Covid-19 testing: Strategies to streamline SARS-CoV-2 molecular assays

N Panpradist, Q Wang, PS Ruth, JH Kotnik, AK Oreskovic, A Miller, ...

EBioMedicine 64, 2021

Aquarium: open-source laboratory software for design, execution and data management

J Vrana, O de Lange, Y Yang, G Newman, A Saleem, A Miller, C Cordray, ...

Synthetic Biology 6 (1), ysab006, 2021

Acoustic measures for real-time voice coaching

Y Li, A Miller, A Liu, K Coburn, LJ Salazar

Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2020

Fast Symbolic Integer-Linear Spectra

J Luntzel, A Miller

arXiv:2410.09053 [math.RA]

Independent Projects

Polymerase Chain Reaction Batching Optimizer, Ruby Gem

Published a clustering algorithm in Ruby for grouping PCRs with similar reaction conditions, leveraging the gradient feature of some thermocyclers to achieve an efficiency gain of up to 96X over the standard laboratory protocol for reactions with known conditions.

Scratch Paper Notes, Mobile App

Created a mobile notepad app featuring a unique user experience. Maintained for 6 years with hundreds of regular users. Recently rebuilt in react native for cross-platform access.

Blood Alcohol Tracker, Mobile App

Developed a mobile app to track a users blood alcohol content live through a drinking session, using the known pharmacokinetics of the absorption and metabolism of ethanol. Built in Java for Android.